

(*) Derivation of Mathematical Model

We derive EOM for 3 subsystems:-

1. Husky Base
2. KUKA Arm
3. Sample Block

1. Husky Base

- Step 1: Identify Forces

From FBD

- $F_{N1}, F_{N2} \rightarrow$ Normal Forces (upward)
- $F_{g-base} \rightarrow$ Weight (downward)
- $F_f \rightarrow$ Friction (opposes motion)
- $a \rightarrow$ Acceleration (forward)

Step 2: Vertical Direction (No motion)

$$\sum F_y = 0,$$

$$F_{N1} + F_{N2} - F_{g-base} = 0$$

$$F_{N1} + F_{N2} = (M_b + M_{arm} + M_s)g$$

Step 3: Horizontal Motion (Newton's 2nd Law)

$$\sum F_x = ma$$

or

$$\sum F_x = M_b a$$

$$\text{Driving Force} = F_L + F_R$$

$$\text{Friction} = F_f$$

$$M_b \dot{v} = F_L + F_R - F_f - F_f$$

$$M_b \dot{v} = F_L + F_R - 2F_f$$

Using viscous model:

$$F_f = \frac{c_r v}{2}$$

So,

$$M_b \dot{v} = F_L + F_R - c_r v$$

Step 4: Rotational Motion (Yaw)

Taking moment about center:

$$\sum M = I_b \dot{\omega}$$

$$I_b \dot{\omega} = b F_R - b F_L - c_{yaw} \omega$$

Step 5: Kinematics

$$\dot{x} = v \cos \theta, \quad \dot{y} = v \sin \theta, \quad \dot{\theta} = \omega$$

Final Base EOM

$$\dot{v} = \frac{F_L + F_R - c_r v}{M_b}$$

$$\dot{\omega} = \frac{b(F_R - F_L) - c_{yaw} \omega}{I_b}$$

2- KUKA Arm:

Step 2: Generalized coordinates

$$q = [q_1, q_2, q_3]$$

Step 2: position of centers of Mass

$$r_{c1} = [L_1 \cos q_1, L_1 \sin q_1]$$
$$r_{c2} = [L_1 \cos q_1 + L_2 \cos(q_1 + q_2),$$
$$L_1 \sin q_1 + L_2 \sin(q_1 + q_2)]$$

Step 3: Velocity

$$v = \frac{dr}{dt}$$

Step 4: Kinetic Energy

$$\dot{T} = \frac{1}{2} \dot{q}^T M(q) \dot{q}$$

Step 5: Potential energy

$$V = (m_1 L_1 + m_2 L_1 + m_{arm} L_1) g \sin q_1$$
$$+ (m_2 L_2 + m_{arm} L_2) g \sin(q_1 + q_2)$$
$$+ m_{arm} L_3 g \sin(q_1 + q_2 + q_3)$$

Step 6: Lagrangian

$$L = T - V$$

1. Step 1: Apply Euler-Lagrange

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = \tau$$

Final Arm Eqⁿ

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau$$

Explicit form

$$\ddot{q} = M^{-1}(q) [\tau - C(q, \dot{q})\dot{q} - G(q)]$$

3. Sample Block

Step 1: Forces

- $F_{\text{constraint}} \rightarrow$ upward
- $F_{g\text{-sample}} \rightarrow$ downward

Step 2: Apply Newton's Law

$$\sum F = ma$$

$$F_{\text{constraint}} - m_s g = m_s a_{\text{acc}}$$

Final eqⁿ

$$a_{\text{acc}} = J(q)\ddot{q} + \dot{J}(q)\dot{q}$$